

- **Useful exergy demand** – Assumed proportional to GDP (1 billion euros → 1000 TJ).
- **Energy system** – Detailed allocation of final exergy by sector, carrier, and end-use, with end-use efficiencies.
- **Renewables** – Solar, onshore/offshore wind, hydro, batteries, hydrogen (electrolysis and fuel cells). Capacity factors and costs evolve over time.
- **CO₂ emissions** – Calculated from final exergy by carrier, with natural gas backup for unmet renewable electricity.
- **Capital & labour** – Sectoral capital stocks, investments, depreciation; labour from working-age population, participation, unemployment, and hours worked.
- **Two scenarios** – BAU (historical trends extended) and Decarbonisation (National Roadmap 2050 targets).

Model Structure (Sheets)

| Sheet | Content |

|-----|-----|

| `Aggregate\_Efficiency\_BAU` | Historical (1960-2022) and projected BAU (2023-2050) shares of final exergy by sector, carrier, end-use, and their efficiencies. |

| `BAU` | Full BAU scenario: renewable capacities, investment, depreciation, labour, TFP, GDP, useful/final exergy, CO₂ emissions. |

| `Aggregate\_Efficiency\_Decarboniz` | Same structure as the BAU sheet, but with input shares following the Portuguese Carbon Neutrality Roadmap 2050 (RNC2050). |

| `Decarbonizatio` | Full Decarbonisation scenario, including extra renewable capacities and costs relative to BAU. |

| `Charts` | Key indicators (efficiency, CO₂, final energy, GDP, renewable GW, sector shares) for both scenarios, ready for plotting. |

Input Data (user-adjustable)

The model is calibrated for **Portugal** but can be adapted for other countries by replacing the input data in the following rows:

| Input | Location (examples) | Description |

|-----|-----|-----|

| **Renewable capacities (GW)** | `BAU!C10:C15`, `Decarbonizatio!C10:C15` | Solar, onshore/offshore wind, batteries, hydrogen, hydro. |

| **Costs per GW** | `BAU!C18:C24`, `Decarbonizatio!C18:C24` | Installation costs for each renewable technology. |

| **Demographics** | `BAU!C55:C59`, `Decarbonizatio!C62:C65` | Working-age population, LFPR, unemployment rate, average hours worked. |

| **Sectoral shares of final exergy** | `Aggregate\_Efficiency\_BAU!C5:C10`, `Aggregate\_Efficiency\_Decarboniz!C5:C10` | Allocation to industry, transport, services, residential, agriculture, energy own use. |

| **Carrier & end-use shares** | `Aggregate\_Efficiency\_BAU!C19:C24`, etc. | Within each sector, shares of electricity, coal, oil, natural gas, renewables, heat, hydrogen, and end-use types (HTH, LTH, etc.). |

| **End-use efficiencies** | `Aggregate\_Efficiency\_BAU!C350:C391`, etc. | Conversion efficiencies for each carrier/end-use pair. |

| **Emission factors** | `BAU!C471:C475` | CO₂ per TJ for coal, oil, natural gas, renewables/waste, heat. |

| **GDP historical** | `BAU!C63:H63` | Actual GDP (M€) used to calibrate the TFP-efficiency relationship and useful exergy intensity. |

Outputs (main results)

All outputs are calculated automatically. Key indicators are shown in the `Charts` sheet for both scenarios.

| Output | Location | Description |

|-----|-----|-----|

| Aggregate efficiency | `BAU!I393:AJ393` (and similar in other sheets) | Final-to-useful exergy efficiency ($\varepsilon = X_{\text{useful}} / X_{\text{final}}$). |

| TFP | `BAU!I61:AJ61` | Total Factor Productivity computed from ε . |

| GDP | `BAU!I63:AJ63` | Gross Domestic Product (M€). |

| Useful exergy | `BAU!I71:AJ71` | X_{useful} (TJ) = GDP $\times$ 1000. |

| Final exergy | `BAU!I72:AJ72` | X_{final} (TJ) = $X_{\text{useful}} / \varepsilon$. |

| Final exergy by carrier | `BAU!C464:AJ468` | Coal/coke, oil, natural gas, renewables/waste, heat, electricity. |

| CO₂ emissions | `BAU!C486:AJ486` | Total CO₂ (Mt). |

| CO₂ intensity | `BAU!C487:AJ487` | CO₂ / GDP (t/€). |

| Renewable capacity | `BAU!I16:AJ16`, `Decarbonizatio!I16:AJ16` | Total installed renewable power (GW). |

| Extra investment cost | `Decarbonizatio!C38:AJ38` | Additional investment (M€) for renewables in the Decarbonisation scenario. |

How to Use

1. **Open `MODEL\_IEEM.xlsx` in Excel** (Macros not required, only formulas).
2. **Review/adjust input data** (see tables above) for the country and scenarios of interest.
3. **The BAU and Decarbonisation scenarios are pre-calculated** – changes to inputs automatically update all dependent sheets.
4. **Results** can be read directly from the `BAU`, `Decarbonizatio`, or `Charts` sheets.
5. **To extend to another country** – replace historical GDP, demographic, capital, and energy allocation data; re-calibrate the TFP-efficiency equation and the useful exergy-GDP coefficient if necessary.

Comments & Notes (as found in the Excel file)

The Excel file contains inline comments that explain the purpose of each sheet and column:

- **`1\_Standard\_Template` (if present)** – “Standard Mode — provide all macro + exergy columns. The kit computes TFP from scratch using factor shares, labour, capital, and GDP. Yellow = optional (used if available; analysis proceeds without them).”

- **`2\_Preloaded\_Template`** – “Pre-loaded Mode — use this sheet if TFP and efficiency series are already available ... Series will be normalized to $z = \ln(x / x_0) + 1$ if not already done.”

- In the `BAU` sheet, the top rows include:

“Scenarios - BAU 2025-2050” and column headers such as “New Solar power [GW]”, “Cost per GW”, “Investment Industry [Mrd Euros]”, “Labour [Hours worked by engaged people]”, etc. These clarify the units and derivation.

- In the `Decarbonizatio` sheet:

“Scenarios up to 2050 - National Roadmap 2050” and “Extra ... compared to BAU” rows show the additional deployment and cost relative to the baseline.

Dependencies

- **Microsoft Excel** (any version that supports standard formulas, no VBA required).
- No external data files – all input data are embedded in the workbook.

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Contact

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